ONE-PAGE BRIEF

Post-quantum cryptography review: rust-bitcoin @ f127ba2ae14e

Client: rust-bitcoin — public repository (passive review) · Source: github.com/rust-bitcoin/rust-bitcoin, commit `f127ba2ae14e` · Reviewed: 2026-11-04 19:18 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **4** Medium
0 Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium	QSI-2026-12578	**Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (7 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12579	**Quantum-vulnerable ECDSA: transaction and message signing (8 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12580	**Quantum-vulnerable ECDSA: role not determined (4 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium	QSI-2026-12581	**Quantum-vulnerable Schnorr (BIP-340): role not determined (3 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Schnorr (BIP-340)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	ECDSA, Schnorr (BIP-340)	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Static review of source code and manifests; files analysed: 125. No requests were sent to the owner's systems.

Full report, content SHA-256: `85f793c3cefd18fd504e7c243ffa6c191c6676f381d6308495e23b444ca7d76c`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.